

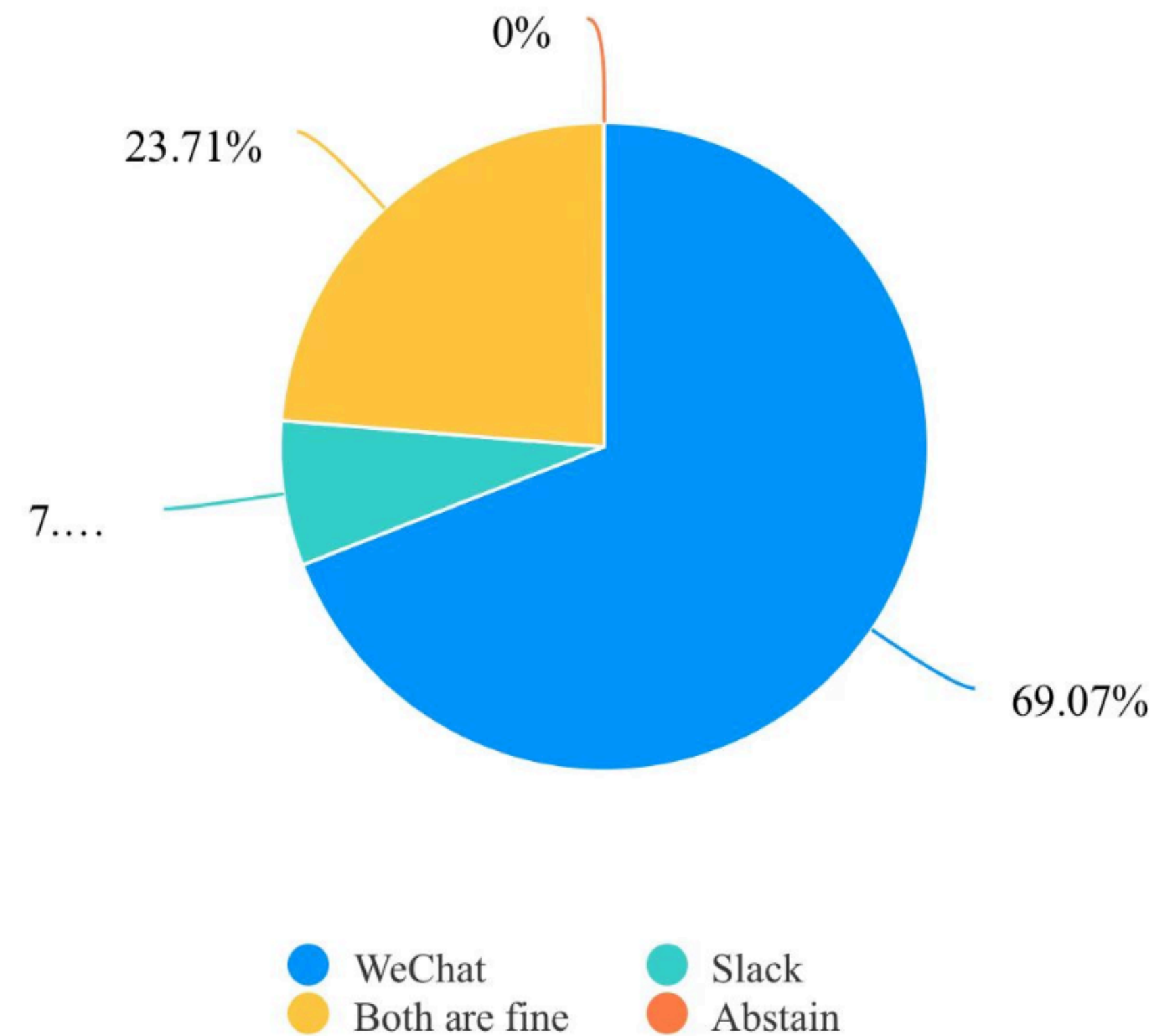


香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

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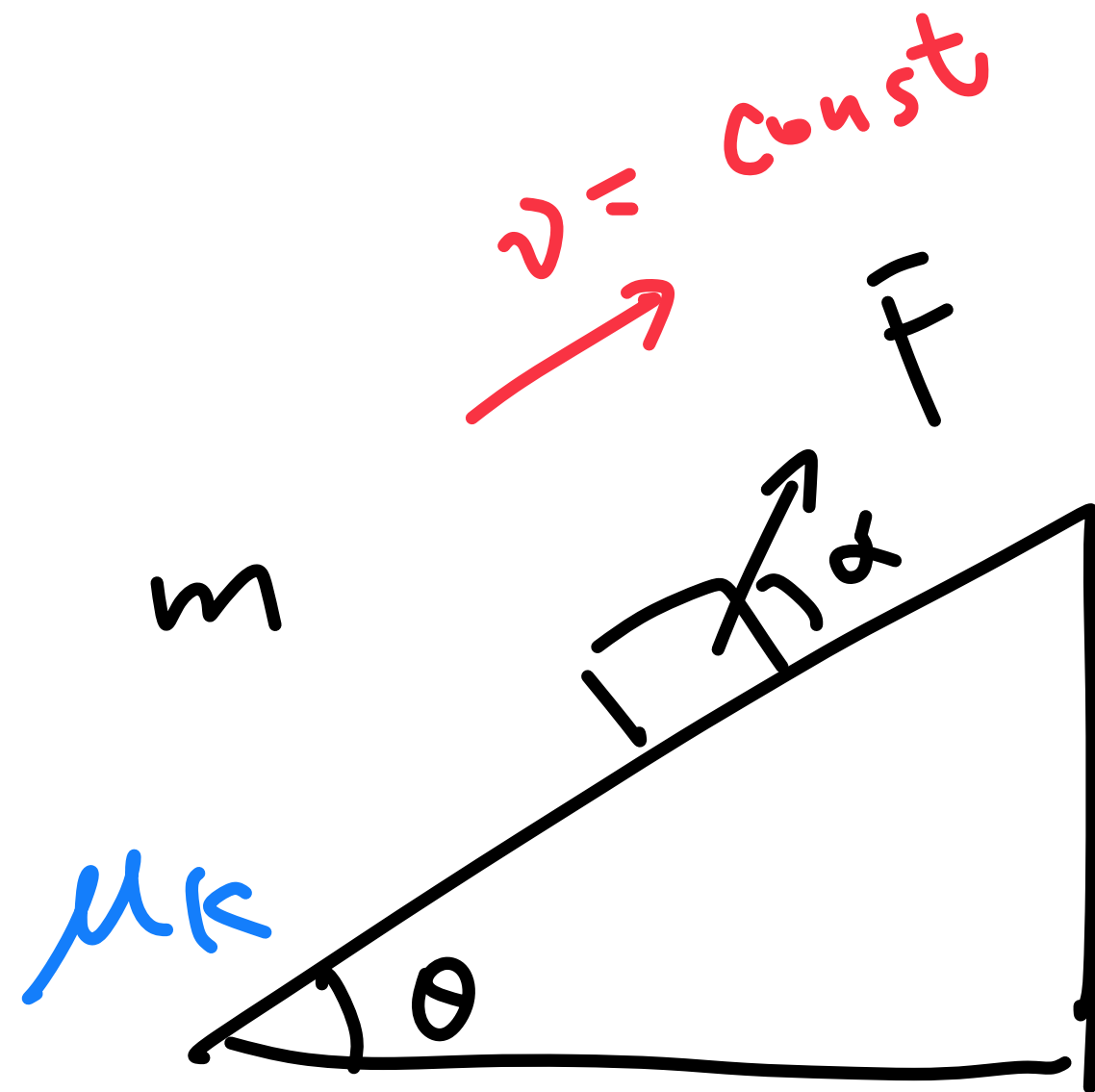
WeChat vs. Slack



- More than half of the students voted. WeChat won the majority (simple majority).
- A WeChat group will be built after the Chinese New Year.

Exercise

e) Following d), at which angle α does force F achieve the minimum?



$$F = \frac{\mu_k mg \cos \theta + mg \sin \theta}{\mu_k \sin \alpha + \cos \alpha}$$

$$\text{max } D = \mu_k \sin \alpha + \cos \alpha$$

$$\frac{dD}{d\alpha} = \mu_k \cos \alpha - \sin \alpha = 0$$

$$\boxed{\mu_k = \tan \alpha}$$

$$\frac{d^2D}{d\alpha^2} = -\mu_k \sin \alpha - \cos \alpha < 0$$

Drag Force

- Drag force is the resistance of a body in liquid fluid (liquid and gas)
- Friction is independent of velocity, but drag force depends on velocity.
- Drag force $f = \underline{k}v$ (small velocity) and $f = \underline{C}v^2$ (large velocity)

f and k are coefficient depending on many properties

$$f_k = \mu_k N$$

Drag Force

- For an object falling in a ^{fluid} liquid with drag force $f = kv$, terminal velocity $v_f = \frac{mg}{k}$

math
①

$$mg - kv = m \frac{dv}{dt} \quad (2nd \text{ law})$$

$$\frac{k}{m} \left(\frac{m}{k} g - v \right) dt = dv$$

$$\int_{t_0}^t -\frac{k}{m} dt = \int_{v_0}^v \frac{dv - \frac{m}{k} g}{v - \frac{m}{k} g}$$

$$-\frac{k}{m}(t - t_0) = \ln \left(\frac{\frac{m}{k} g - v}{\frac{m}{k} g - v_0} \right)$$

$$= \ln \frac{\frac{m}{k} g - v}{\frac{m}{k} g - v_0}$$

$$\frac{mg}{k} - v = \left(\frac{mg}{k} - v_0 \right) e^{-\frac{k}{m}(t-t_0)}$$

$$v = \frac{mg}{k} - \left(\frac{mg}{k} - v_0 \right) e^{-\frac{k}{m}(t-t_0)}$$

$t \rightarrow +\infty$

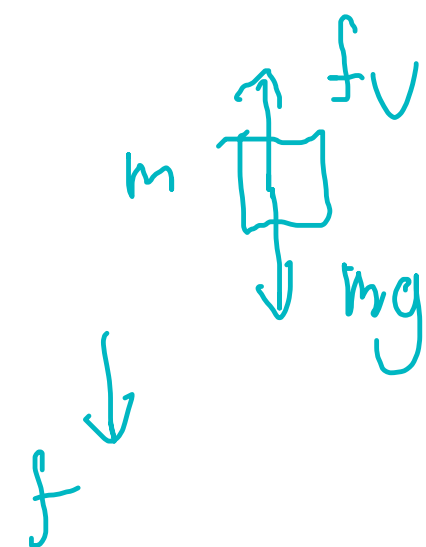
$$V_f = \frac{mg}{k}$$

Physics
②

At terminal state

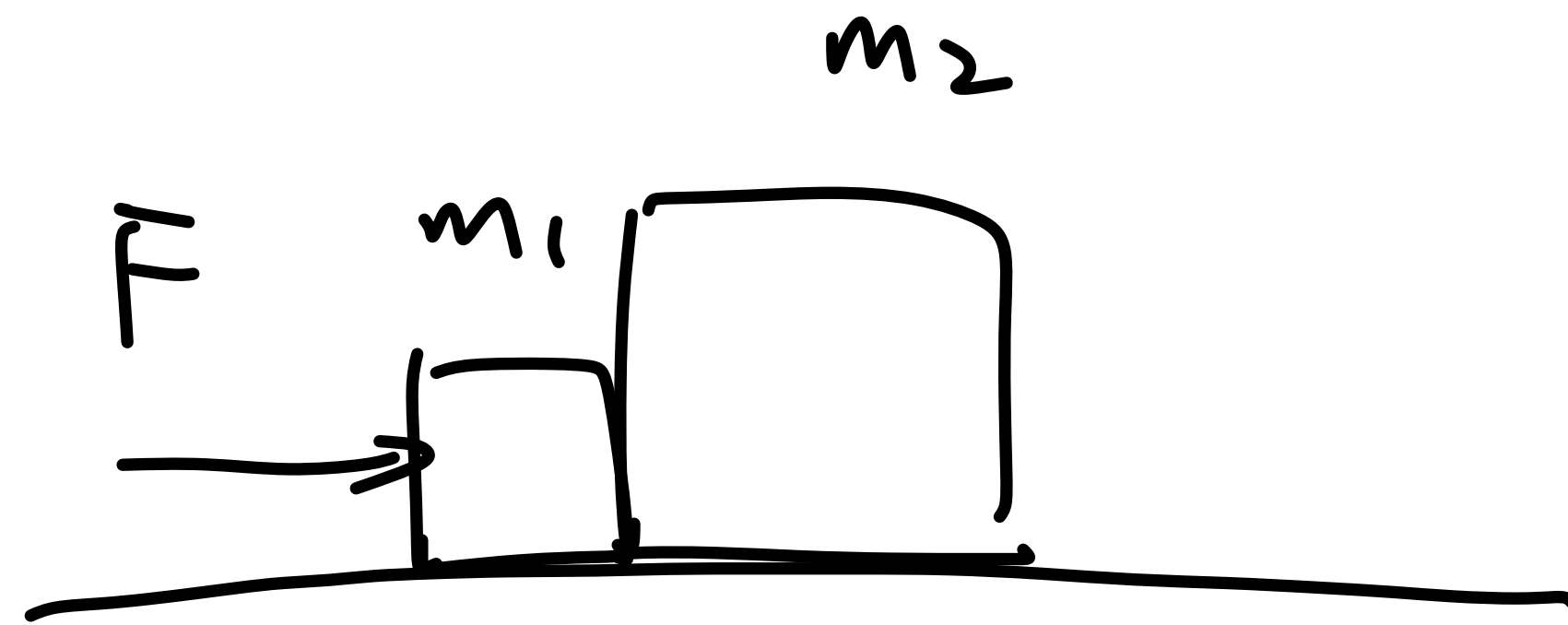
$$kV_f = mg$$

$$V_f = \frac{mg}{k}$$



$$\int \frac{1}{x} dx = \ln x$$

System of Bodies



Find force & acceleration of m_2

$$m_1 : F - F_1 = m_1 a_1$$

$$m_2 : F_2 = m_2 a_2$$

$$F' = F_1 = F_2 \text{ (3rd law)}$$

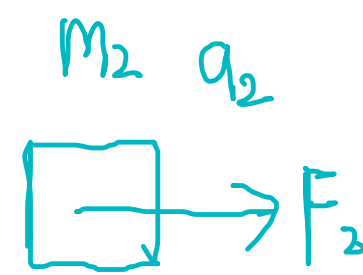
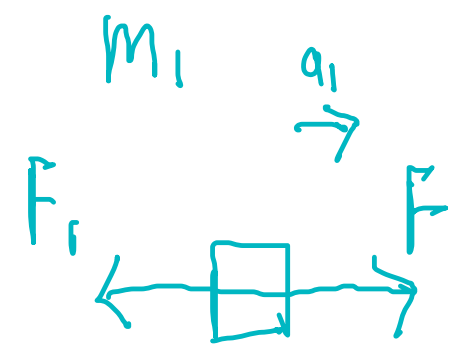
$$a = a_1 = a_2$$

$$F - F' = m_1 a \quad (1)$$

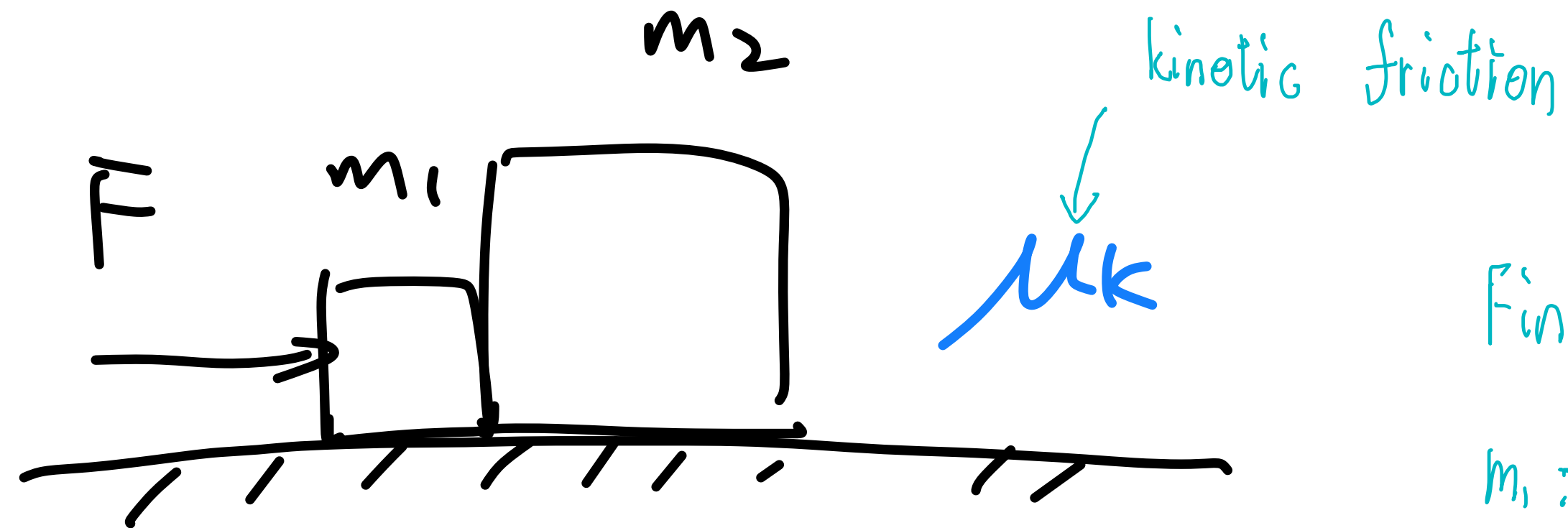
$$F' = m_2 a \quad (2)$$

(1) + (2)

$$a = \frac{F}{m_1 + m_2}$$



System of Bodies



Find f & a of m_2

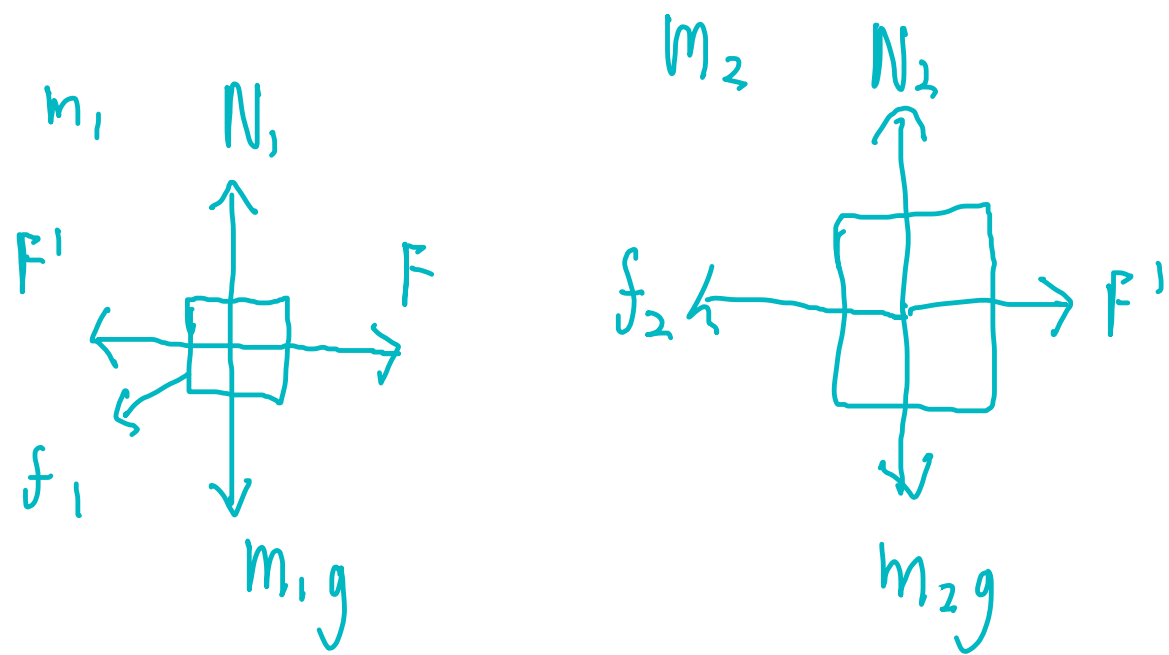
$$m_1: F - F' - \mu_k m_1 g = m_1 a \quad (1)$$

$$m_2: F' - \mu_k m_2 g = m_2 a \quad (2)$$

$$(1) + (2) \quad F - \mu_k (m_1 + m_2) g = (m_1 + m_2) a$$

$$a = \frac{F}{m_1 + m_2} - \mu_k g$$

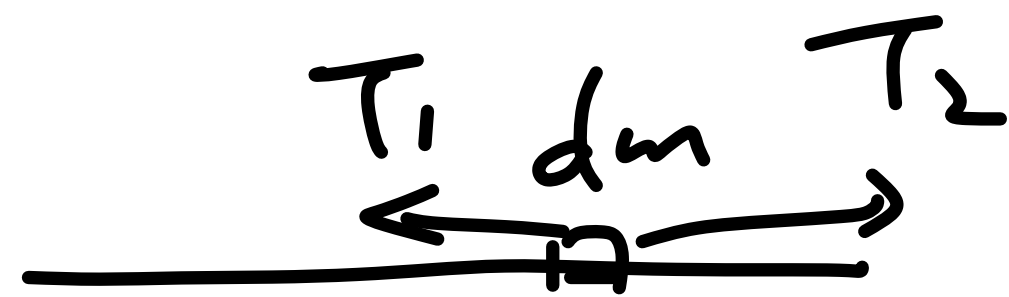
$$\begin{aligned} F' &= m_2 a + \mu_k m_2 g \\ &= \left(\frac{F}{m_1 + m_2} - \mu_k g \right) m_2 + \mu_k m_2 g \\ &= \frac{m_2}{m_1 + m_2} F \end{aligned}$$



Tension

- Tension is the force in a wire/rope, denoted by T
- Two assumptions
 - Massless (indicated by a thin wire, massless rope, etc.): tension is the same in the wire/rope

↑ Important assumption

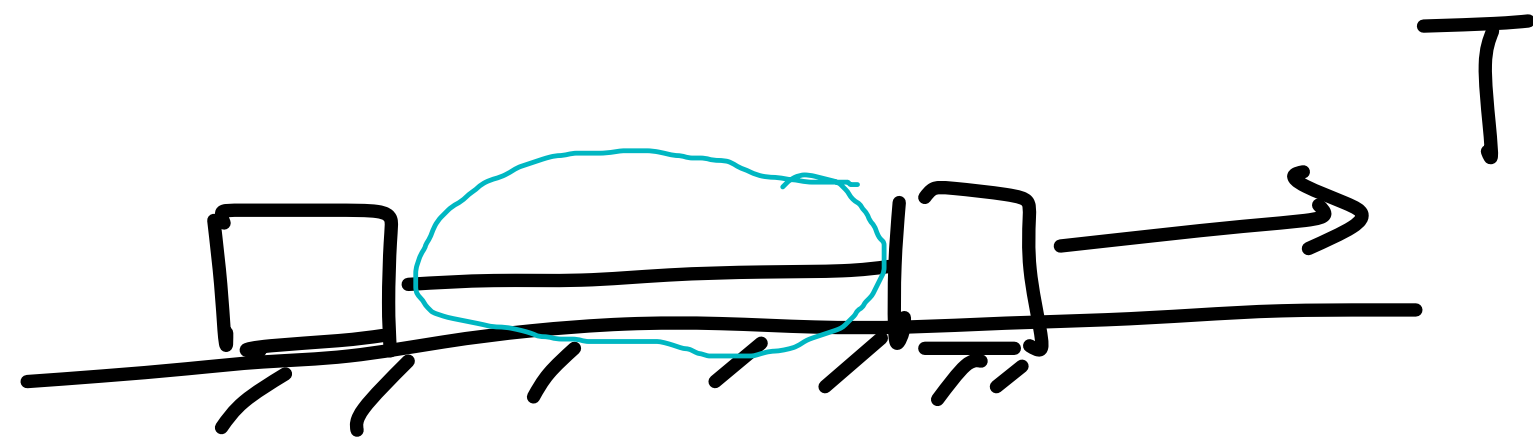


$$T_2 - T_1 = dm \cdot a$$
$$= 0$$

$T_1 = T_2$ at any place on the rope

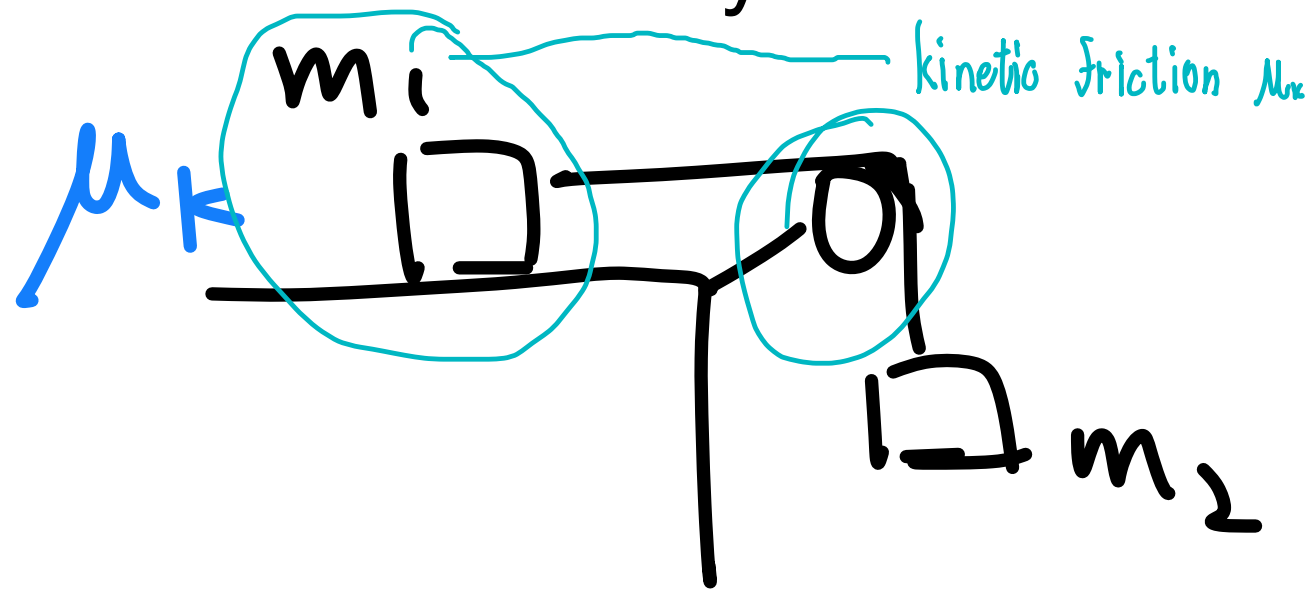
Tension

- Tension is the force in a wire/rope, denoted by T
- Two assumptions
 - Massless (indicated by a thin wire, massless rope, etc.): tension is the same in the wire/rope
 - Fixed length: bodies attached by the same wire/rope have the **same** motion, i.e. **same** displacement, velocity, and acceleration.



Pulley

- Wire/rope with a pull changes the direction of tension, otherwise the same assumptions as wire/rope
- Usually assumed to be smooth (get rid of friction)



direction is same

acceleration, magnitude same

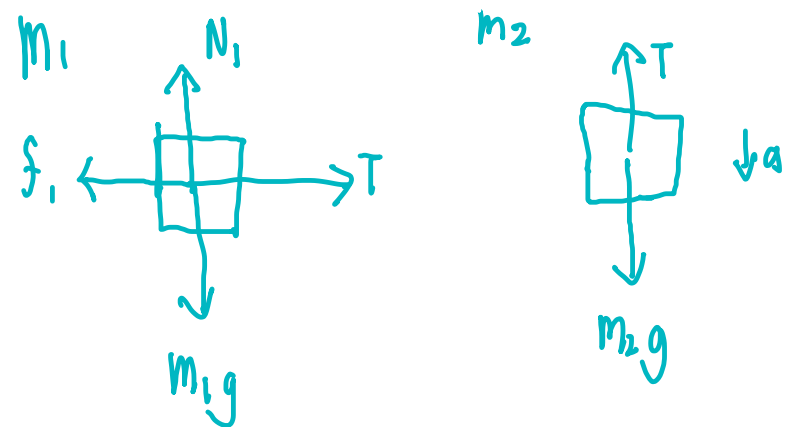
find acceleration

$$m_1: T - \mu_k m_1 g = m_1 a \quad (1)$$

$$m_2: m_2 g - T = m_2 a \quad (2)$$

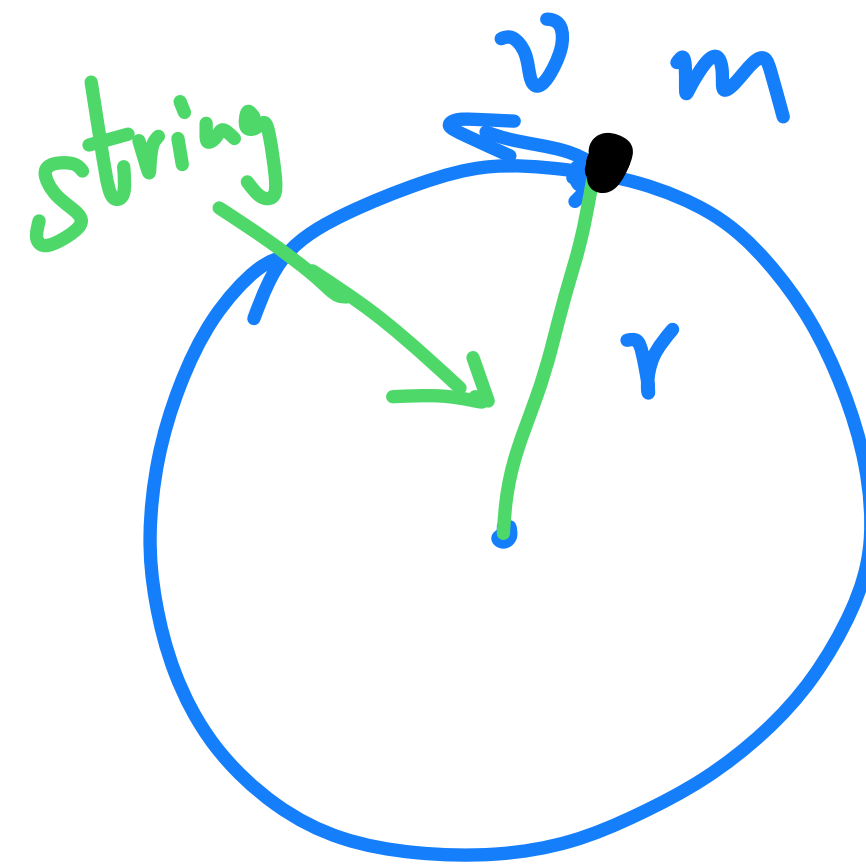
$$(1) + (2) \quad m_2 g - \mu_k m_1 g = (m_1 + m_2) a$$

$$a = \frac{m_2 g - \mu_k m_1 g}{m_1 + m_2}$$



Centripetal Force

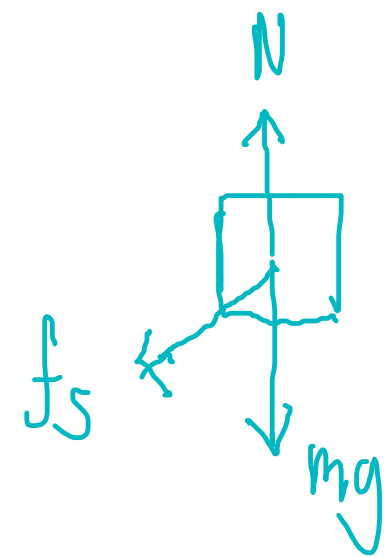
- Centripetal force maintains uniform circular motion, $F = m \frac{v^2}{r}$
- Many forces can play the role of centripetal force!



Centripetal Force

a) A car with mass m is circling on horizontal ground with a radius of r . The ground has a coefficient of **static** friction μ_s . Find the maximum velocity v_{max} of the car.

radius r

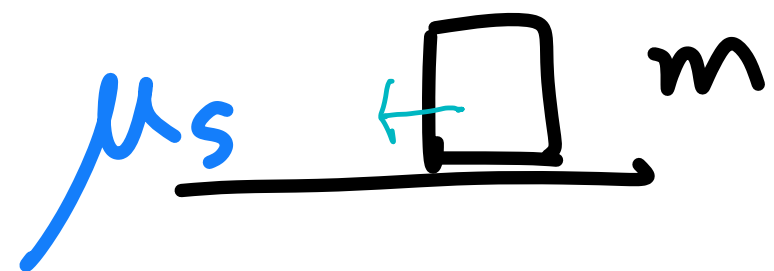


$$f_s = m \frac{v^2}{r}$$

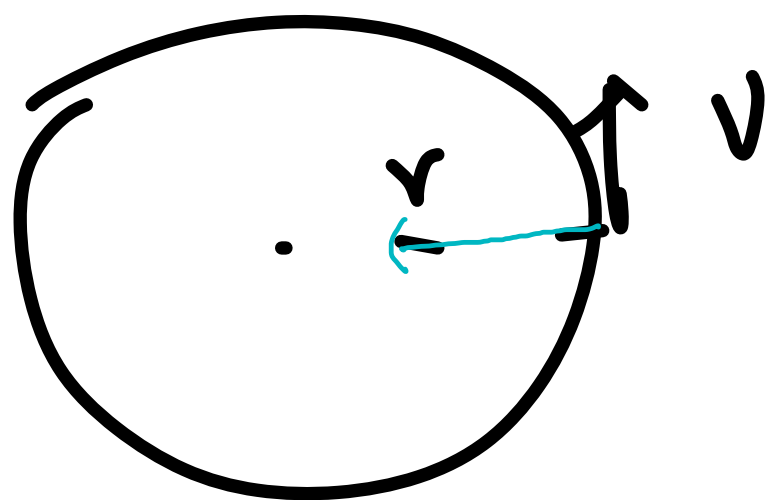
$$\leq f_{s, \max} = \mu_s mg$$

$$m \frac{v^2}{r} \leq \mu_s mg$$

$$v \leq \sqrt{\mu_s g r}$$

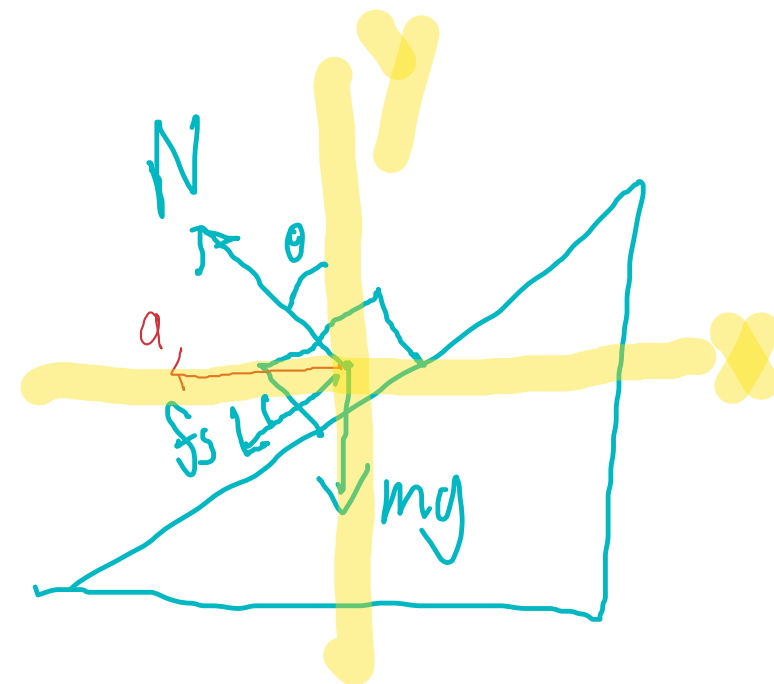
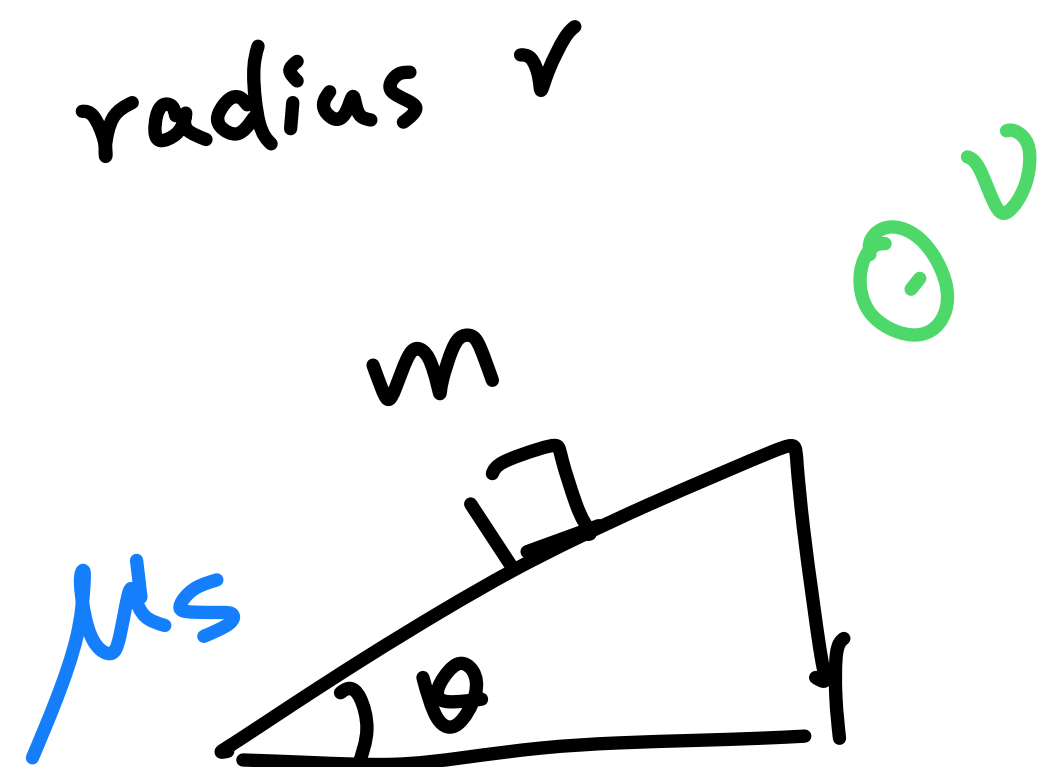


Bird view



Centripetal Force

b) Following a), the car is now at a ramp with an angle θ to the horizontal plane. Find the maximum velocity v_{max} of the car.



$$y: N \cos \theta - f_s \sin \theta - mg = 0 \quad (1)$$

$$x: N \sin \theta + f_s \cos \theta = m \frac{v^2}{r} \quad (2)$$

$$f_s \leq \mu_s N \quad (3)$$

$$(1) \rightarrow (3) \quad N \cos \theta - mg \leq \mu_s N \sin \theta$$

$$N \leq \frac{mg}{\cos \theta - \mu_s \sin \theta} \quad (4)$$

$$(2) \quad m \frac{v^2}{r} \leq N \sin \theta + \mu_s N \cos \theta$$

$$= N (\sin \theta + \mu_s \cos \theta)$$

$$= \frac{mg}{\cos \theta - \mu_s \sin \theta} (\sin \theta + \mu_s \cos \theta)$$

$$m \frac{v^2}{r} \leq \frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} mg$$

$$v \leq \sqrt{\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta} \cdot gr}$$

can be infinity

$$\mu_s = 0$$

$$v \leq \sqrt{\tan \theta \cdot gr}$$